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ENT 4501

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Zotero-Ready Bibliography (WearableTech-AV)

Microcontroller & Firmware Documentation

Espressif Systems. ESP32 Technical Reference Manual. <https://www.espressif.com>

Espressif Systems. ESP-IDF Programming Guide. <https://docs.espressif.com/projects/esp-idf>

Arduino. Arduino Core for ESP32 Documentation. <https://docs.arduino.cc>

Audio, I2S & FFT Resources

Kosme. arduinoFFT Library. <https://github.com/kosme/arduinoFFT>

DFRobot. Fermion I2S MEMS Microphone (INMP441) Wiki. <https://wiki.dfrobot.com>

SparkFun Electronics. Understanding FFT for Sound Analysis. <https://learn.sparkfun.com>

Paillet, M. ESP32 I2S Microphone Tutorial. <https://medium.com>

Electronics & Components

Adafruit Industries. WS2812B RGB LED Guide & Datasheet. <https://learn.adafruit.com/adafruit-neopixel-uberguide>

Adafruit. NeoPixel Reference Documentation. <https://learn.adafruit.com>

SparkFun Electronics. Potentiometer Basics. <https://learn.sparkfun.com/tutorials/potentiometers>

Vishay Semiconductors. IR Receiver Modules Datasheet. <https://www.vishay.com>

PCB & KiCad Documentation

KiCad Project. KiCad Documentation (v8/v9). <https://docs.kicad.org>

Digi-Key Electronics. ESP32-DevKitC-V1E Product Page. <https://www.digikey.com>

PCBWay. How to Export Gerber Files (KiCad). <https://www.pcbway.com>

OSH Park. KiCad PCB Design Guidelines. <https://docs.oshpark.com>

3D Modeling & Fabrication

Autodesk. Fusion 360 Documentation. <https://help.autodesk.com>

Ultimaker. Cura 3D Printing Software Documentation. <https://ultimaker.com/software/ultimaker-cura>

MatterHackers. Designing 3D Printed Enclosures. <https://www.matterhackers.com>

Interaction Design, Serial, p5.js

Processing Foundation. p5.js Reference. <https://p5js.org/reference>

Lauren McCarthy et al. p5.js Sound Library Documentation. <https://p5js.org>

Processing Foundation. Serial Communication with p5.js. <https://creative-coding.de-contextualize.com/serial/>

Coursework & Example Projects

City Tech E-Media Technology Dept. MTEC 3525 / 4501 Course Materials.

Random Nerd Tutorials. ESP32 Audio Spectrum Projects. <https://randomnerdtutorials.com>

Various creators. DIY LED Music Visualizer Projects (YouTube).

Datasheets (Used Directly)

Espressif Systems. ESP32-WROVER-IE Datasheet. <https://www.espressif.com>

TDK Invensense. INMP441 Microphone Datasheet. <https://invensense.tdk.com>

WorldSemi. WS2812B LED Datasheet. <https://cdn-shop.adafruit.com/datasheets/WS2812B.pdf>

Bourns. Rotary Potentiometer Datasheet. <https://www.bourns.com>

Power & Charging Modules

DFRobot. USB-C Lithium Battery Charging Module Documentation. <https://wiki.dfrobot.com>

Texas Instruments. Li-Ion Battery Management Application Notes. <https://www.ti.com>